

Hi-TGNet: Hierarchical Tumor Guidance Network for Explainable Brain Tumor Classification

Souhardo Rahman¹, Md. Nasif Safwan¹, Taharat Muhammad Jabir¹, Mahamodul Hasan Mahadi¹, MD Iftekharul Mobin^{1,*}, M. Abdullah-Al-Wadud², and Jia Uddin^{3,*}

¹Department of Computer Science, American International University-Bangladesh, Dhaka, Bangladesh

²Department of Software Engineering, College of Computer and Information Sciences, King Saud University, Riyadh 11543, Saudi Arabia

³AI and Big Data Department, Woosong University, Daejeon, South Korea

*Corresponding author: MD Iftekharul Mobin (e-mail: iftekhar.mobin@aiub.edu) and Jia Uddin (jia.uddin@wsu.ac.kr)

ABSTRACT

Automated brain tumor classification assists time-sensitive and individualized treatment planning, yet most deep learning solutions exhibit a pragmatic compromise between quality, efficiency, and strength throughout heterogeneous MRI protocols. Vision Transformers are capable of capturing global data, but they are frequently too expensive to run in real-time in clinical settings. Our architecture Hi-TGNet is a lightweight framework that maintains multi-scale discriminability and enhances cross-domain reliability in the case of 4-class brain tumor diagnosis. Hi-TGNet is based on an Adaptive Spiral Block, which extends the effective receptive field using parallel dilated branches, a Hierarchical Tumor Guidance module, which focuses attention on tumor-relevant areas to make more discriminative representations, and an uncertainty-aware dual-head predictor, which helps to scale decisions on ambiguous cases. In order to facilitate interpretability, we give explainability analysis through Grad-CAM++ and saliency-based attribution to confirm that predictions are guided by anatomically significant areas instead of background effects. We adopt a two-stage training protocol, performing slice-level pretraining on multi-modal BraTS 2020 and fine-tuning on the Brain MRI ND-5 dataset. Hi-TGNet achieves 97.32% accuracy on the ND-5 held-out test set and maintains strong performance under distribution shift, reaching 98.70% on BRISC 2025 and 93.54% on the Mendeley Brain Tumor MRI dataset using the same inference protocol without dataset-specific tuning. With only 3.98M parameters, Hi-TGNet delivers an effective balance of accuracy, efficiency, and external robustness, supporting scalable deployment for automated neuro-oncology screening and decision support.

Keywords: Brain tumor classification, Guided attention, Convolutional neural network

Introduction

The effects of brain tumor on world health is a burning issue to contemporary medicine. Such malignancies have been linked to high mortality rates as well as high neurological disability among a wide range of patients. Current clinical statistics released in 2025 point to the fact that the number of primary brain tumors is on the increase at the same time¹. The seriousness of this load can be explained by the fact that recent epidemiological statistics indicate that meningiomas are the most common primary intracranial tumor, occupying 42.6% of the diagnostics, and gliomas occupy 22.2% of the diagnoses but are the majority of the malignant pathologies². The prognosis in each of the subtypes is vastly different with statistics indicating that glioblastoma, the most aggressive type of glioma, has a median survival rate of only 8 months and a five-year relative survival rate of about 6.9% compared to pituitary tumors, which constitute about 15-17% of all primary brain tumors and has lower mortality but high long-term endocrine morbidity^{2,3}. The future of neuro-oncology is in the establishment of the personalized treatment plans, which are based on the accurate and fast tumor type classification^{4,5}. The ability to distinguish Glioma, Meningioma, and Pituitary tumors effectively is critical to deciding the most effective surgical and therapeutic procedures to apply to each patient⁶.

Deep learning and machine learning have completely transformed the interpretation of medical imaging and data analysis by clinicians⁷⁻⁹. These superior technologies enable automated analysis of MRI and CT images at a degree of precision never before achievable¹⁰. Machine learning models are able to detect complex structure patterns and minor anomalies that may go unnoticed when being reviewed manually. This change has shifted the radiology profession into a more trustworthy and efficient diagnostic system in which artificial intelligence is an important decision support tool⁶.

The existing state of brain tumor classification is characterized by the extensive number of advanced neural network architectures. Convolutional Neural Networks and Vision Transformers are among the methods that researchers largely resort to

in order to detect the brain tumors with high accuracy in multiclass detection tasks^{11,12}. Hybrid models that utilize a traditional network spatial feature extraction method combined with the global context of the transformer blocks are the focus of many studies nowadays¹⁰. Moreover, the use of transfer learning has become a standard practice for adapting models trained on general image datasets to the specific requirements of medical diagnostics¹³. Most recent standards indicate that such methods are able to achieve high measures of performance on standardized clinical data.

Although automated brain tumor classification has progressed tremendously, there are still a number of research gaps that restrict the clinical applicability of most automated brain tumor models. The existing architectures are usually too complicated and computationally intricate, limiting real-time applications in clinical applications¹⁴. Also, medical images are noisy by nature, and the model may fail to distinguish between the tumor and the background, and instead give a false positive result, leading to diagnostic ambiguity¹⁵. There are also no studies conducted in a structured manner on the effects of data augmentation levels and specific attentional mechanisms on model stability and generalization in the medical context¹⁶.

The conventional CNNs use a fixed-size convolutional kernel that has a limited receptive field that is not sufficient to detect fine-grained tumor edges and global contextual features of the image as a whole¹⁷. The size and shape of tumors change drastically and the properties of single scales might be unable to represent such a change. The high-level layers receive semantic features yet lose spatial resolution, even though low-level CNN layers receive high-resolution positional features which are prone to noise¹⁸. Existing methods have challenges with inability to capture such multi-scale characteristics.

Moreover, many top-performing models have low interpretability and comprehension of the features that form the basis of their forecasts¹⁹. Models which not only achieve the best accuracy, but also have transparency, that is, by showing salient regions that indicate the presence of the tumor, are badly required, to ensure that classification is performed depending on the pertinent tumor characteristics, and not on artifacts.

We suggest Hi-TGNet as a solution to these problems. This custom architecture is built on a Spiral CNN backbone designed for both efficiency and high precision. We incorporate multi-scale feature extraction and a novel multi-stage attention framework to improve classification accuracy and interpretability.

The key contributions of this work include the following:

- A lightweight Hi-TGNet encoder with stagewise context modeling and cross-scale fusion for 4-class brain tumor classification.
- An Adaptive Spiral Block that captures multi-scale context using parallel depthwise branches with learnable dilation and efficient fusion.
- A Hierarchical Tumor Guidance module that produces multi-level attention and a spatial guidance map to promote pathology-aware refinement.
- An uncertainty-aware dual-head predictor that outputs both class logits and per-class uncertainty to handle ambiguous cases.
- A two-stage training and evaluation protocol with BraTS 2020 slice-level pretraining, ND-5 fine-tuning, and external validation on BRISC 2025 and the Brain Tumor MRI dataset.

Our research contributes to the study landscape by providing a modular and interpretable framework for neuro-imaging analysis. We demonstrate how specialized attention and structured training strategies can overcome the limitations of standard deep learning models in the field of oncology.

Related Work

Automated classification of brain tumors has been evolving faster, as it is being shifted away to more advanced deep learning paradigms, rather than the classical machine learning approaches. In this section, the recent developments will be discussed, with the emphasis on the shift to the Transformer hybrids, the incorporation of attention mechanisms, and the role of XAI in explaining and justifying architecture decisions.

Early Stages of Automated Detection

Prior to the reign of end-to-end deep learning, automated detection had mostly been based on manual feature extraction with classical classifiers. Puttegowda et al. reexamined these traditional approaches and established that although such algorithms as Support Vector Machines (SVM) and Random Forests were capable of reaching reasonable accuracy of up to 96% on balanced datasets, they needed a lot of preprocessing and were not able to learn hierarchical features based on raw MRI data alone²⁰. Alshomrani also stresses that due to the semantic gap between low-level pixel intensities and high-level tumor pathology, these inception phases were bottlenecked and forced the transition to deep learning to overcome this limitation¹⁹. Dhole et

al. present an in-depth overview of this shift, indicating that although the first Neural Networks were still a major step in the right direction, they were frequently susceptible to overfitting, especially when they were fed on small medical datasets²¹. In addition, comparison studies by Chen et al. revealed that initial convolutional models such as regular ResNet50 were frequently incapable of capturing the fine-grained boundary information of gliomas without synthetic data augmentation or an elaborate post-processing step²².

CNNs, Transformers, and Hybrid Architectures

The area has recently divided into the developed Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs). Comparative analysis on the BRISC dataset in 2025 showed that the heavy CNNs such as ResNet18 achieved the same accuracy as the hybrid MobileViT models but provided smaller inference time, making it difficult to assume that the Transformer is universally optimal in all clinical tasks²³. Nevertheless, pure Transformer methods are on the increase. Recent research proposed an end-to-end Vision transformer that achieved 99.64% accuracy stating that global receptive field of ViTs effectively reduces the inductive bias of CNNs including translation equivariance which can be harmful when dealing with irregular tumor shapes²⁴. On the other hand, hybrid architectures are designed in order to merge the two worlds. In a study, Swin Transformers were compared with regular CNNs and it was discovered that Swin-Large has a top accuracy of 99.37%, but the computation expense is a weakness of the actual implementation of Swin Transformers in more practical real-time applications²⁵. In the same vein, a 2025 study suggested that EfficientNetV2 be paired with ViTs, and the geometric mean of the results be maximized to increase precision, recall, and F1-score, which addresses the local-feature limitations of pure Transformers²⁶. Zhang et al. also addressed this gap with ETUnet, introducing Transformer modules to the bottleneck of U-Net-based architectures to improve the ability of global dependencies in the model without degrading the spatial resolution of convolutional encoders²⁷. Alongside these supervised trends, self-supervised learning (SSL) is increasingly used in brain tumor classification to reduce reliance on scarce labels. In our prior work, we developed and evaluated an SSL-based pretraining strategy and found that it strengthens downstream classification performance²⁸.

Ensemble, Federated Models, and XAI

Recent research has shifted the focus of literature to Federated Learning (FL) and Explainable AI (XAI) to resolve the problem of data privacy and model reliability. A 2025 study proposed an interpretable FL framework using GoogleNet, allowing institutions to collaboratively train tumor classifiers without sharing sensitive patient data, achieving high accuracy while maintaining privacy standards²⁹. Complementing this, hybrid ensemble models have shown superior robustness where a study introduced a model fusing Deep Learning with XGBoost and Grad-CAM, achieving 99.77% accuracy. More importantly, the presence of XAI (Grad-CAM) enabled clinicians to visualize the decision-making process, confirming the fact that the model was oriented towards the tumor core instead of background artifacts³⁰. Continuing the FL trend, Dai et al. reviewed privacy preserving methods, and found that by incorporating ResNet50 into federated averages (FedAvg), it was possible to robustly classify meningioma and pituitary tumors on heterogeneous datasets³¹. It has been demonstrated that new geometric mean ensembles stabilize predictions, decreasing the variance commonly associated with single-model deployments²⁷.

Lastly, critical validation of design decisions of complexity is being demanded in high-impact work to an increased degree. Ablation is defined by Sheikholeslami et al. as the systematic extraction of parts of the system to measure their respective contributions, and address the distinction between what actually drives performance in the system and what is unnecessary complexity³². Historical medical imaging experiments have demonstrated that bigger models are not necessarily superior, that simpler backbones can perform equally on larger clinical datasets when the data and preprocessing are robust to them, and that extra complicated attention designs can overfit smaller clinical datasets^{23,33}.

Attention Mechanisms and Specialized Convolutions

To overcome the drawbacks of regular convolutions, more recent studies have looked at specialized kernels and attention modules. The recent efforts presented Channel-Attention DenseNet with the assistance of dilated convolutions. These enlarged kernels did not add any parameters to the receptive field, but still enabled the model to jointly capture the mass of tumor and the surrounding edema, which is very similar to the idea of Spiral convolutions that we have proposed in our work³⁴. Similarly, a fetal imaging study demonstrated that attention-guided convolution could dynamically redistribute feature weights in the suppression of noise, a fact that can be immediately extended to the background-suppression of MRI images³⁵. The efficacy of Convolutional Block Attention Module (CBAM) was also verified as introducing the module to YOLOv7 improved the detection of small pituitary tumors by refining both channel and spatial dimension feature maps in a more effective way than without it³⁶. Moreover, hybrid CNN-LSTM-Attention models have been used to represent sequential dependencies in MRI slices where it was proven that attention mechanisms are fundamental to current analysis of medical images³⁷.

Critical Gaps and Limitations

Despite these achievements, some remarkable challenges still exist. Alshomrani cites the black box property of deep learning as a longstanding obstacle to clinical adoption, and says that interpretability of models is typically negatively linked to complexity¹⁹. Another 2025 report further points out that most state-of-the-art systems have been found to be sensitive to dataset imbalance and overlap where recall tends to lag behind precision of visually similar types of tumors³⁸. Lastly, the heterogeneity of MRI acquisition protocols presents a generalization gap where models learned on standardized data sets such as BraTS tend to fail when inputted with raw clinical data with motion artifact or differing contrast values, a shortcoming that is not fully overcome by the existing domain adaptation methods to date³⁹.

Synthesis of Research Trends and Study Motivation

Throughout the 2022 to 2025 research landscape, a consistent trade-off in automated brain tumor detection is apparent. Transformer-based models, and hybrid architectures address both the inadequate global context modeling and minimized the locality bias of their predecessors, but this can be very costly in terms of computation and memory, so it can be costly to apply in clinical systems. This pressure drives the motivation to develop a model lightweight enough to be inferred quickly and yet being able to model multi-scale and long-range visual cues needed to make reliable decisions on a tumor.

Hi-TGNet is motivated by this gap. It does not rely on a heavy global attention backbone but uses multi-scale spiral blocks with parallel dilated depthwise separable convolutions to expand the effective receptive field of a convolutional design. This offers more overarching coverage of context and maintains the skeletal footprint of parameters. Moreover, the model incorporates explicitly regularized mechanisms of attention that are learned throughout training to ensure that responses of attention are more accurately associated with tumors and repressed when using healthy scans. Lastly, our experimental design focuses on interpretability and causal attribution with the contribution of each architectural component and training signal being isolable and optimized with controlled evaluations and external testing, responding to the practical reasons behind reliability and transparency that still cause slow clinical adoption.

Methodology

Hi-TGNet is a small but powerful CNN that we suggest in classifying brain tumors at two modes of operation. The network is instantiated with C output classes: $C=2$ and $C=4$ instantiations of the same architecture are used to learn slice-level tumor presence and the Brain MRI ND-5 diagnostic task respectively. The model is based on a stagewise encoder architecture that gradually adds channel capacity and spatial resolution, followed by consolidation of the multi-scale evidence through cross-scale fusion, and ends with the generation of both class logits and uncertainty estimate by using two parallel heads of the prediction module. In the network, we explicitly learn the context at many scales of receptive fields and produce a guidance map, which drives features to focus on pathologic areas of interest.

Hi-TGNet Pipeline Overview

Figure 1 summarizes the end-to-end dataflow. Given an input image $\mathbf{I} \in \mathbb{R}^{3 \times 224 \times 224}$, the network produces a prediction vector $\mathbf{y} \in \mathbb{R}^C$ and an uncertainty vector $\mathbf{u} \in \mathbb{R}^C$. Let $\mathcal{E}(\cdot)$ denote the stagewise encoder, $\mathcal{F}(\cdot)$ the cross-scale fusion, and $\mathcal{H}(\cdot)$ the dual-head predictor. The forward pass can be written compactly as:

$$\begin{aligned} \{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \mathbf{f}_4\} &= \mathcal{E}(\mathbf{I}), \\ \{\mathbf{f}'_1, \mathbf{f}'_2, \mathbf{f}'_3, \mathbf{f}'_4\} &= \mathcal{F}(\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \mathbf{f}_4), \\ \mathbf{z} &= \text{DualPool}(\mathbf{f}'_4), \\ (\mathbf{y}, \mathbf{u}) &= \mathcal{H}(\mathbf{z}), \end{aligned} \tag{1}$$

where $\mathbf{z}$ is a compact descriptor formed from the final fused stage. The network is used for 4-way classification on ND-5, which contains 17,888 labeled MRI images distributed across glioma, meningioma, pituitary, and healthy classes.

Stagewise Context Encoder

The encoder follows a four-stage hierarchy with a lightweight stem and three strided downsampling transitions. With base width $c_0=48$, the channel dimensions per stage are:

$$(c_1, c_2, c_3, c_4) = (48, 96, 144, 192), \tag{2}$$

while spatial resolutions follow $224 \rightarrow 112 \rightarrow 56 \rightarrow 28 \rightarrow 14$ due to the stem stride and subsequent stride-2 transitions. Each stage contains blocks of context and an explicit guidance unit which produces refined features as well as a guidance map at the stage level. Indicating the feature of stage s by $\mathbf{x}_s$, the encoder carries out:

$$\mathbf{x}_{s+1} = \text{Down}_s(\Phi_s(\mathbf{x}_s)), \tag{3}$$

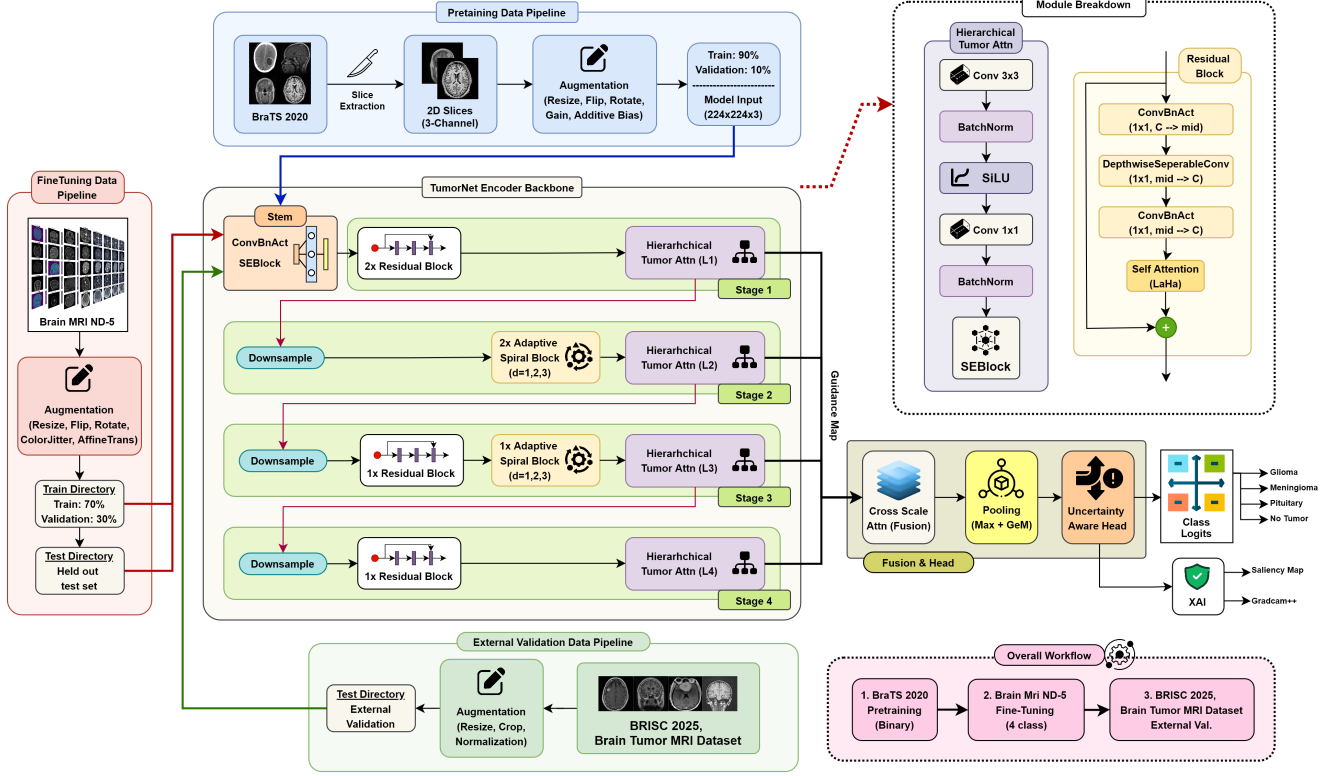


Figure 1. Proposed Hi-TGNet pipeline overview. Multi-resolution features and guidance maps are generated by the stagewise encoder, evidence is consolidated across different stages by cross-scale fusion and a dual-head predictor generates both class logits and uncertainty.

where $\Phi_s(\cdot)$ stacks local context blocks and a guidance module, and $\text{Down}_s(\cdot)$ is a stride-2 convolutional transition. This hierarchy balances local texture extraction at high resolution with semantic abstraction at lower resolution, which is critical for tumor subtype discrimination where both boundary detail and global context can be informative.

Spiral Residual Context Blocks

Hi-TGNet employs two context operators which are complementary to each other. The former is a residual refinement block, which jointly relies on pointwise expansion, separable depthwise convolution, and a light attention gate to enhance local stability. The second is the Adaptive Spiral Block as shown in Figure 2 which is aimed to increase the effective receptive field without heavy self-attention cost. Given an input tensor $\mathbf{x} \in \mathbb{R}^{B \times C \times H \times W}$ with C divisible by 3, we split channels into three groups $\mathbf{x} = [\mathbf{x}^{(1)}; \mathbf{x}^{(2)}; \mathbf{x}^{(3)}]$ and apply depthwise convolutions with learnable effective dilations:

$$\mathbf{y}^{(i)} = \text{DWConv}(\mathbf{x}^{(i)}; d^{(i)}), \quad i \in \{1, 2, 3\}. \quad (4)$$

Since standard convolution is constrained by integer dilation, we approximate continuous dilation by interpolating between adjacent integer dilation values. All the three outputs of the branches are then concatenated and mixed with a pointwise projection and recalibrated by an Squeeze-and-Excitation gate. Lastly, a residual connection is used to stabilize the optimization and retain a gradient flow. This design yields multi-scale context sensitivity inside a single block, which is especially helpful when tumor appearance varies in size and texture across slices.

Hierarchical Tumor Guidance

To make feature refinement explicitly pathology-aware, each stage includes a Hierarchical Tumor Attention unit that generates both feature reweighting and a spatial guidance map, as shown in Figure 3. Let $\mathbf{x} \in \mathbb{R}^{B \times C \times H \times W}$ be the stage feature. We compute L parallel level features $\{\mathbf{h}_\ell\}_{\ell=1}^L$ using lightweight depthwise and pointwise filtering. These are fused to form a reference tensor $\mathbf{f}$, from which level weights are predicted:

$$\pi = \text{softmax}(\text{Conv}(\text{GAP}(\mathbf{f}))) \in \mathbb{R}^{B \times L}. \quad (5)$$

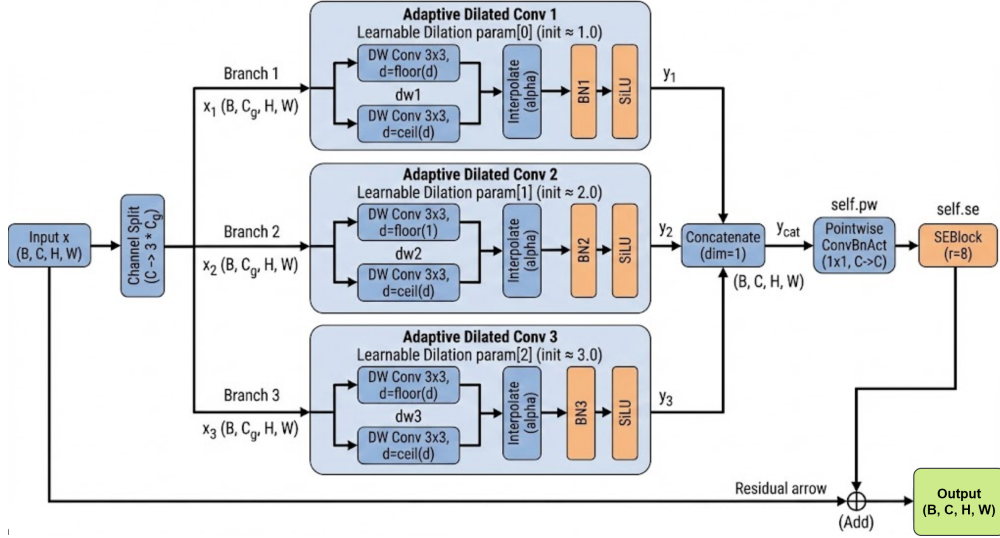


Figure 2. Adaptive Spiral Block. The channels are divided into three parallel branches, which are processed through learnable effective dilations, followed by the fusing pointwise mixing and SE recalibration of the channels on a residual path.

We predict level weights by applying global average pooling to the fused feature tensor, followed by a 1×1 convolution and a softmax across the L levels. These weights are used to compute a level-weighted mixture of the multi-level features. A lightweight convolutional guidance head then produces a spatial guidance map that gates the refinement, so the final output is formed by adding a guidance-modulated residual update to the input feature. Guidance maps are produced at multiple stages and later aggregated after spatial alignment, enabling the model to emphasize tumor-relevant regions consistently across resolutions.

Scale Fusion and Dual Head

After the four encoder stages, we consolidate information across resolutions using a Cross-Scale Attention fusion module. Each stage feature is projected to a unified embedding dimension of $d=128$ using a 1×1 convolution and resized to match the deepest stage resolution:

$$\tilde{\mathbf{f}}_s = \text{Resize}(\text{Proj}_s(\mathbf{f}_s)), \quad s \in \{1, 2, 3, 4\}. \quad (6)$$

The projected features are fused through lightweight refinement and re-injected back into each stage using stage-specific refinement with a learnable mixing gate.

For prediction, we construct a compact descriptor from the final fused stage feature using dual pooling:

$$\mathbf{z} = \left[\text{GeM}(\mathbf{f}'_4); \text{MaxPool}(\mathbf{f}'_4) \right]. \quad (7)$$

The descriptor is passed to a dual-head prediction module that outputs class logits and a per-class uncertainty estimate. The classification branch uses a learned gating mechanism to modulate the final hidden representation before logits are produced, while the uncertainty branch applies a Softplus activation to ensure non-negative uncertainty values, as shown in Figure 4.

Experiments

Datasets and Evaluation Protocol

We evaluate our approach using a two-stage training setup (pretraining then fine-tuning) followed by both in-domain testing and external validation. Throughout all experiments, we use a fixed class order with four categories: `glioma_tumor`, `meningioma_tumor`, `no_tumor`, and `pituitary_tumor`. This fixed ordering is used consistently when computing metrics, constructing confusion matrices, and mapping dataset folder names to class indices.

BraTS (pretraining source). We initialize our network from a checkpoint obtained via pretraining on the BraTS 2020 dataset, which provides multi-institutional pre-operative glioma MRI with strong acquisition variability^{40–42}. The BraTS 2020 training cohort contains 369 subjects spanning high-grade glioma and low-grade glioma cases, and each subject includes

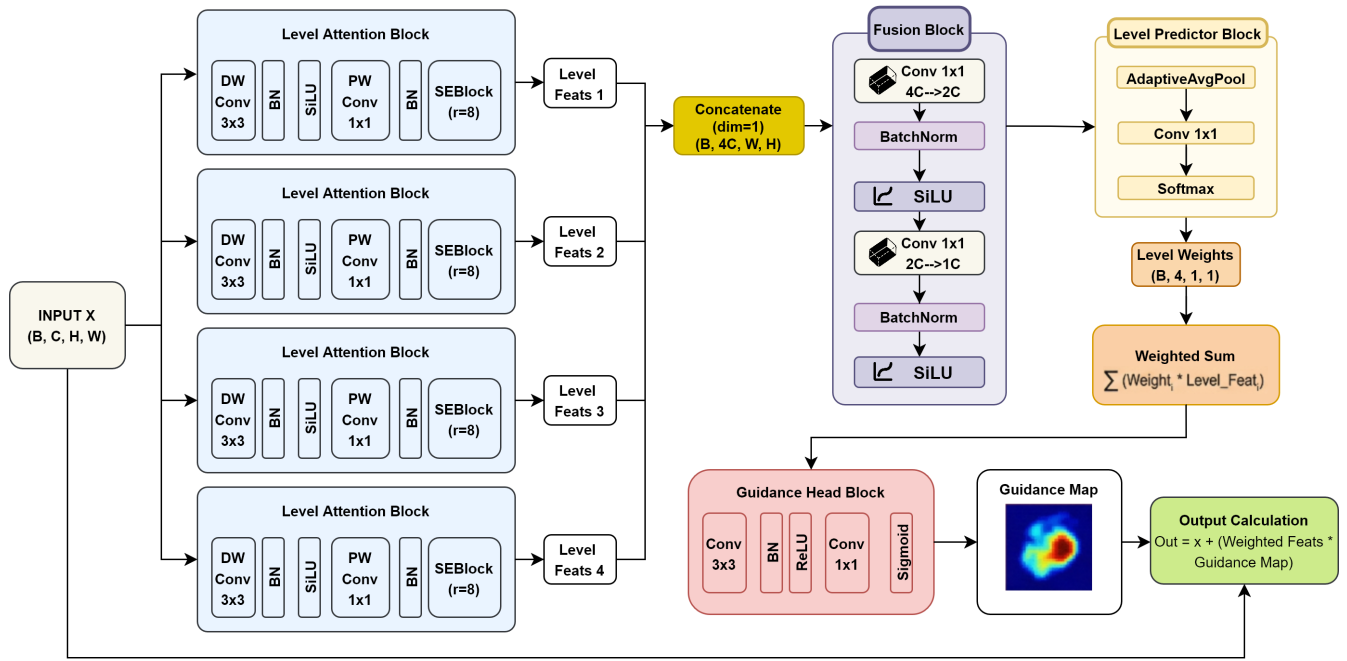


Figure 3. Hierarchical Tumor Guidance. Multiple level attentions are fused, level weights are predicted, and a spatial guidance map gates the residual refinement of stage features.

four co-registered MRI modalities: T1, T1ce post-contrast, T2, and FLAIR. BraTS also provides expert annotations for tumor subregions, commonly evaluated as whole tumor, tumor core, and enhancing tumor, which makes it useful for learning tumor-aware representations during pretraining. We used 90% of cases for training and 10% for validation. In our pipeline, BraTS is used only to learn transferable weights, while ND-5 supplies the supervised signal for the final 4-way classification task.

Brain MRI ND-5 (primary fine-tuning dataset). Our main supervised dataset is a publicly available brain tumor classification corpus containing 17,888 labeled 2D images grouped into four diagnostic categories: glioma, meningioma, pituitary, and healthy (no tumor)⁴³. The images include common anatomical planes (axial, sagittal, coronal). The dataset is organized into two top-level directories: *Training* and *Testing*. We use:

- **Training:** split into 70:30 ratio where first portion is used for model fine-tuning and the second is kept as a validation set.
- **Testing:** kept reserved and used only for final performance evaluation

This protocol ensures that hyperparameter selection is separated from the final test evaluation on ND-5.

BRISC 2025 (external validation). To evaluate the distribution shift-resistance, we provide external validation of the BRISC 2025 test split, which is a curated dataset to design the brain tumor segmentation and classification task⁴⁴. BRISC 2025 has 6,000 contrast-enhanced T1-weighted MRIs of four subtypes: glioma, glioma, meningioma, no_tumor, and pituitary. The samples were taken along axial, sagittal, and coronal planes. In addition to image-level labels, BRISC provides expert annotations in the form of segmentation masks, which makes it a strong benchmark for evaluating whether a classifier remains reliable under changes in acquisition style and tumor appearance. Consistent with our external validation protocol, no tuning, threshold selection, or augmentation changes are performed on BRISC, and the model is evaluated strictly in an inference-only setting.

Brain Tumor MRI Dataset from Mendeley (external validation). We also report external results on an additional hospital dataset organized in class-specific folders and referenced in the code as *Epic* and *CSCR hospital Dataset/Test*⁴⁵. This dataset contains 12,064 pre-processed T1-weighted contrast-enhanced MRI brain images spanning four categories: glioma_tumor, meningioma_tumor, pituitary_tumor, and no_tumor. It is used only for evaluation and is processed using the same deterministic inference pipeline as BRISC.

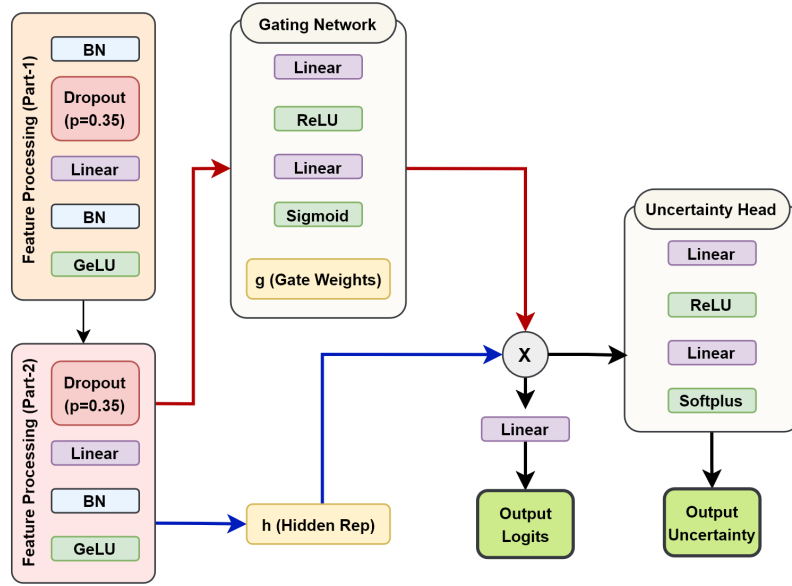


Figure 4. Dual-head predictor. A gated classification branch outputs logits, while a parallel uncertainty branch outputs non-negative per-class uncertainty using a Softplus activation.

Preprocessing and Data Augmentation

BraTS slice preprocessing and augmentation for pretraining

For BraTS pretraining, we construct a slice-level binary task that predicts tumor presence from 2D slices. Each training sample is generated by randomly sampling a single axial slice from a 3D case using three modalities `flair`, `t1ce`, and `t2` as three input channels. With probability 0.7, the sampled slice is forced to be tumor-positive when tumor slices exist in the case, otherwise a random slice is selected. Each modality slice is normalized using z-score normalization followed by min-max scaling to $[0, 1]$, and then resized to 224×224 . When enabled, the corresponding segmentation slice is binarized into a tumor mask and resized to the same spatial resolution.

BraTS pretraining uses lightweight, label-preserving augmentations applied consistently to the image and mask. These include random horizontal and vertical flips, random rotations by $k \cdot 90^\circ$ with $k \in \{0, 1, 2, 3\}$, and mild intensity perturbation via a random gain and bias. This design increases robustness to orientation and intensity variation while preserving the anatomical structure of tumor regions.

ND-5 training augmentation for fine-tuning

For ND-5 fine-tuning, we use a single stochastic augmentation pipeline implemented with torchvision transforms. Given an input image $\mathbf{I}$, the augmented sample $\mathbf{I}'$ is produced by random resized cropping to 224 with scale in $[0.75, 1.0]$, random horizontal flipping with probability 0.5, random vertical flipping with probability 0.2, random rotation with a maximum magnitude of 20° , color jitter with parameters $(0.2, 0.2, 0.2, 0.1)$ for brightness, contrast, saturation, and hue, and a random affine transform with translation up to 10% and scale in $[0.9, 1.1]$. The augmented image is then converted to a tensor and normalized using ImageNet statistics. This augmentation pipeline is used only for training on ND-5 and is not applied to validation or test sets.

Deterministic validation and test preprocessing

All validation and test evaluations on the ND-5 held-out test set, BRISC 2025, and Brain Tumor MRI Dataset from Mendeley test sets use a deterministic preprocessing pipeline that resizes the shorter side to 256, applies a center crop of 224×224 , converts the image to a tensor, and normalizes it using ImageNet statistics. Let $\mathbf{t}$ be the tensor after resizing and cropping. Channel-wise normalization is applied as:

$$\begin{aligned}
 x_c &= \frac{t_c - \mu_c}{\sigma_c}, \\
 \mu &= (0.485, 0.456, 0.406), \\
 \sigma &= (0.229, 0.224, 0.225).
 \end{aligned} \tag{8}$$

Model and Training Strategy

Architectural overview

We use a hierarchical encoder with progressive downsampling and channel expansion. The network consists of a feature extraction stem followed by four stages operating at spatial resolutions 112, 56, 28, and 14, with channel widths $48 \rightarrow 96 \rightarrow 144 \rightarrow 192$ for the base setting $\text{base_c}=48$. The stem uses stride-2 convolution for early downsampling and includes an SE block for channel recalibration. Every residual block uses an expand depthwise project structure and uses LaHa attention to promote features. The spiral blocks work along parallel branches within the channel groups to seize the multi-scale context efficiently. A hierarchical attention module also processes each stage output to generate a guidance map that assists not only in inducing the feature modulation but also as a regularization during training. A cross-scale attention module combines multi-resolution features at different stages and then pools and performs the final classification. Table 1 reports the parameter allocation across these components, totaling 3.981M parameters.

Table 1. Parameter breakdown by network component with corresponding tensor shapes.

Layer	Total Parameters	Output Shape
Stem	23,048	(1, 48, 112, 112)
Stage1	132,669	(1, 48, 112, 112)
Stage2	315,899	(1, 96, 56, 56)
Stage3	843,970	(1, 144, 28, 28)
Stage4	1,705,571	(1, 192, 14, 14)
Cross_Scale	142,916	(1, 192, 14, 14)
Pooling	1	(1, 384)
Head	817,160	(1, 4)
TOTAL	3,981,234	–

Channel recalibration with Squeeze-and-Excitation. The network uses Squeeze-and-Excitation blocks to adaptively emphasize informative channels and suppress weak or noisy responses. Given a feature map $\mathbf{F}$, global context is summarized with global average pooling and passed through a lightweight bottleneck to produce channel-wise gates:

$$\mathbf{z} = \sigma(\mathbf{W}_2 \delta(\mathbf{W}_1 \text{GAP}(\mathbf{F}))), \quad (9)$$

where GAP denotes global average pooling, δ is ReLU, and σ is sigmoid. The gates $\mathbf{z}$ rescale $\mathbf{F}$ channel-wise, enabling the model to strengthen subtle tumor cues that may be underrepresented in early feature maps while keeping the added computation small.

LaHa attention in residual blocks. The transformed features have LaHa attention in each residual block to refine them before they are concatenated with the skip connection. The first step in LaHa is to generate a probe-based gate and do a Hadamard modulation followed by Gaussian smoothing to minimize the high-frequency noise in the attention response. Once smoothing is performed, channel and spatial gates are computed in order to emphasize the most informative channels and spatial locations. The channel descriptor in the implementation is based on the spatial mean and standard deviation of the smoothed activation and therefore the gating is less sensitive to differences in MRI image intensities or contrasts among scans. The attention-modulated features are then added back through the residual connection, preserving optimization stability while improving spatial selectivity.

Dual pooling and uncertainty-aware classification. After the final encoder stage, the model performs global aggregation using a dual pooling strategy that concatenates GeM pooling with global max pooling. GeM pooling is defined as:

$$\text{GeM}(\mathbf{F}) = \left(\frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W \mathbf{F}_{i,j}^p \right)^{\frac{1}{p}}, \quad (10)$$

where p is learned and initialized to 3.0. Concatenating GeM and max pooling preserves both distributed context and strong localized responses, which is important when discriminative tumor evidence may occupy a small fraction of the image. The resulting descriptor is passed to an uncertainty-aware gated head that outputs class logits and a non-negative per-class uncertainty estimate, enforced using a Softplus activation, allowing the model to express confidence in addition to the predicted label.

Pretraining then fine-tuning

Training follows a two-stage procedure. First, we pretrain Hi-TGNet on BraTS using a binary objective for tumor presence from slice-level inputs, with optional supervision from segmentation masks via a guidance-mask loss. Pretraining uses AdamW with learning rate 3×10^{-4} , weight decay 10^{-4} , batch size 10, and runs for 10 epochs. The BraTS split uses 90% of cases for training and 10% for validation.

Second, we fine-tune the model on ND-5 for 4-class classification using the ND-5 Training directory, which is split 70:30 using stratified sampling, where 70% is used for training during finetuning and the rest is used as a validation set. The ND-5 Testing directory is reserved as a held-out test set. We initialize the 4-class model from the BraTS checkpoint while reinitializing the classification head by excluding head parameters during weight loading. Fine-tuning uses AdamW with learning rate 2.5×10^{-4} , weight decay 10^{-4} , batch size 12, and runs for 35 epochs. Class weights are computed from the ND-5 training folder using an inverse-frequency scheme and are used to weight the classification loss.

Optimization objective

The fine-tuning objective combines a label-smoothed focal classification loss with two auxiliary regularizers derived from the predicted uncertainty and the learned guidance map. It is a weighted combination of three terms: the primary label-smoothed focal loss reduces overconfidence and emphasizes visually ambiguous samples, the uncertainty term discourages high uncertainty for correct predictions without penalizing genuinely difficult cases, and the guidance-map term encourages tumor-consistent activation, suppresses spurious attention for healthy scans, and promotes spatial coherence through a total-variation penalty. In addition, during BraTS pretraining only, we use an extra mask-guided attention loss when segmentation masks are available to directly align the guidance map with tumor regions, providing a better spatially grounded initialization for ND-5 fine-tuning.

Label-smoothed focal loss with class weighting. Let $\mathbf{p}$ be the softmax probabilities and $\mathbf{y}$ the label-smoothed target distribution with smoothing factor $\varepsilon = 0.1$. We adopt a label-smoothed focal objective to address two practical issues observed in medical image classification. First, even when a dataset is approximately balanced, some categories can be easier to learn, causing training to be dominated by many easy samples. The focal factor $(1 - p_t)^\gamma$ reduces the contribution of confident predictions and forces the optimizer to focus on harder, ambiguous cases where tumors are subtle or visually similar across classes. Second, MRI labels can contain mild noise and borderline cases, so label smoothing reduces overconfident fitting by softening the targets, which improves calibration and generalization. Finally, we include inverse-frequency weights α_k to prevent any residual class skew from biasing the decision boundary. The resulting loss is:

$$\begin{aligned} \mathcal{L}_{\text{cls}} &= \alpha(\mathbf{y}) (1 - p_t)^\gamma \left(- \sum_{k=1}^K y_k \log p_k \right), \\ p_t &= \sum_{k=1}^K y_k p_k, \quad \gamma = 2, \quad \varepsilon = 0.1, \end{aligned} \tag{11}$$

where $\alpha(\mathbf{y}) = \sum_{k=1}^K y_k \alpha_k$ aggregates the class weights α_k according to the smoothed target.

Here, K is the number of classes, y_k is the smoothed target for class k , and p_k is the predicted probability for class k .

Uncertainty regularization. The head predicts per-class uncertainty values $\mathbf{u} \in \mathbb{R}_{\geq 0}^K$. The design goal is to encourage the model to be confident when it is correct, while not forcing low uncertainty on genuinely ambiguous or misclassified samples. For this reason, the uncertainty term is applied only when the prediction matches the ground truth. This avoids a degenerate solution where the network simply drives uncertainty to zero for all inputs and also prevents penalizing uncertainty on hard cases where high uncertainty is desirable. The uncertainty regularizer is defined as:

$$\mathcal{L}_{\text{unc}} = \mathbb{E} [u_{\hat{y}} \mathbb{I}(\hat{y} = y)], \tag{12}$$

where $\hat{y}$ is the predicted class and $\mathbb{I}(\hat{y} = y)$ activates the penalty only for correct predictions.

Guidance regularization from hierarchical attention. The model generates a merged guidance map $g \in [0, 1]^{1 \times H \times W}$, where greater values depict spacial regions which the network focus on. In the case of ND-5, the classes of tumors are defined as $\{0, 1, 3\}$ and healthy as 2. Since ND-5 provides only image-level labels, we use a label-only regularizer to shape g during fine-tuning. In the case of tumor images, the regularizer would promote more high guidance responses by punishing low attention, whereas in healthy images it would punish high guidance. We also add a total-variation smoothness penalty on g to discourage sudden pixel-to-pixel transformation and as a result discourage noisy and speckled attention patches and encourage coherent ones. We also impose a weak global magnitude constraint on the average value of g to eliminate the triviality of the guidance map being saturated.

The full fine-tuning objective is an addition of the main 4-class classification loss with an uncertainty regularizer and guidance regularizer. The classification term optimizes the ND-5 prediction, the uncertainty term discourages excessive uncertainty on correct predictions, and the guidance term enforces class-consistent and spatially smooth attention behavior.

Mask-guided attention loss in BraTS pretraining. BraTS provides pixel-level tumor annotations, which we use to explicitly ground the model’s guidance mechanism during pretraining. The network produces a guidance map $g \in [0, 1]^{1 \times H \times W}$, and we supervise it with the resized binary tumor mask m so that high guidance aligns with true tumor regions and remains low elsewhere. This additional supervision encourages anatomically meaningful attention and provides a better initialization for ND-5 fine-tuning, where only image-level labels are available. We implement this constraint using a binary cross-entropy loss on the guidance logits:

$$\mathcal{L}_{\text{mask}} = \text{BCEWithLogits}\left(\log \frac{g}{1-g}, m\right), \quad (13)$$

and weight it by $\lambda_{\text{mask}} = 0.3$ during BraTS pretraining.

Evaluation Setup

Inference configuration and metrics

All evaluations use a batch size of 32 and the deterministic test preprocessing pipeline described earlier. Inference is executed on P100 GPU. We report multi-class Accuracy, macro-averaged Precision, Recall, and F1, and macro AUC computed in a one-vs-rest manner using softmax probabilities. Performance is also summarized with a 4×4 confusion matrix for each evaluated dataset. Key training and optimization hyperparameters used across BraTS pretraining and ND-5 fine-tuning are summarized in Table 2.

Table 2. Key hyperparameters for the experimental setup.

Parameter	Value
Input size	224×224
Input channels	3
Random seed	42
Optimizer	AdamW
Weight decay	1×10^{-4}
LR scheduler	CosineAnnealingWarmupRestarts
Scheduler params	$T_0=10$, $T_{\text{mult}}=2$, $\eta_{\text{min}}=0.01 \times \text{LR}$
Warmup epochs	2 (BraTS), 3 (ND-5)
BraTS pretrain epochs	10
BraTS batch size	10
BraTS learning rate	3×10^{-4}
BraTS loss	FocalLoss $\gamma=2$, label smoothing 0
Attention-mask loss	enabled, $\lambda=0.30$
ND-5 fine-tune epochs	35
ND-5 batch size	12
ND-5 learning rate	2.5×10^{-4}
ND-5 loss	FocalLoss $\gamma=2$, label smoothing 0.1

Test-time augmentation

The code optionally applies test-time augmentation using horizontal flips, right-angle rotations, and small scale changes. For the final ND-5 held-out test set, we use scales $\{0.95, 1.0, 1.05\}$, horizontal flips in $\{\text{false}, \text{true}\}$, and rotations in $\{0, 90, 180, 270\}$. Predictions from all augmented views are combined using entropy-based weighting, which assigns higher weight to low-entropy predictions.

Results Analysis

In-Domain Evaluation on Brain MRI ND-5

Our proposed Hi-TGNet model performs well on the ND-5 held-out test set in-domain and is able to maintain this reliability on external validation suggesting that the learned representation can be generalized to outside the training distribution. The model

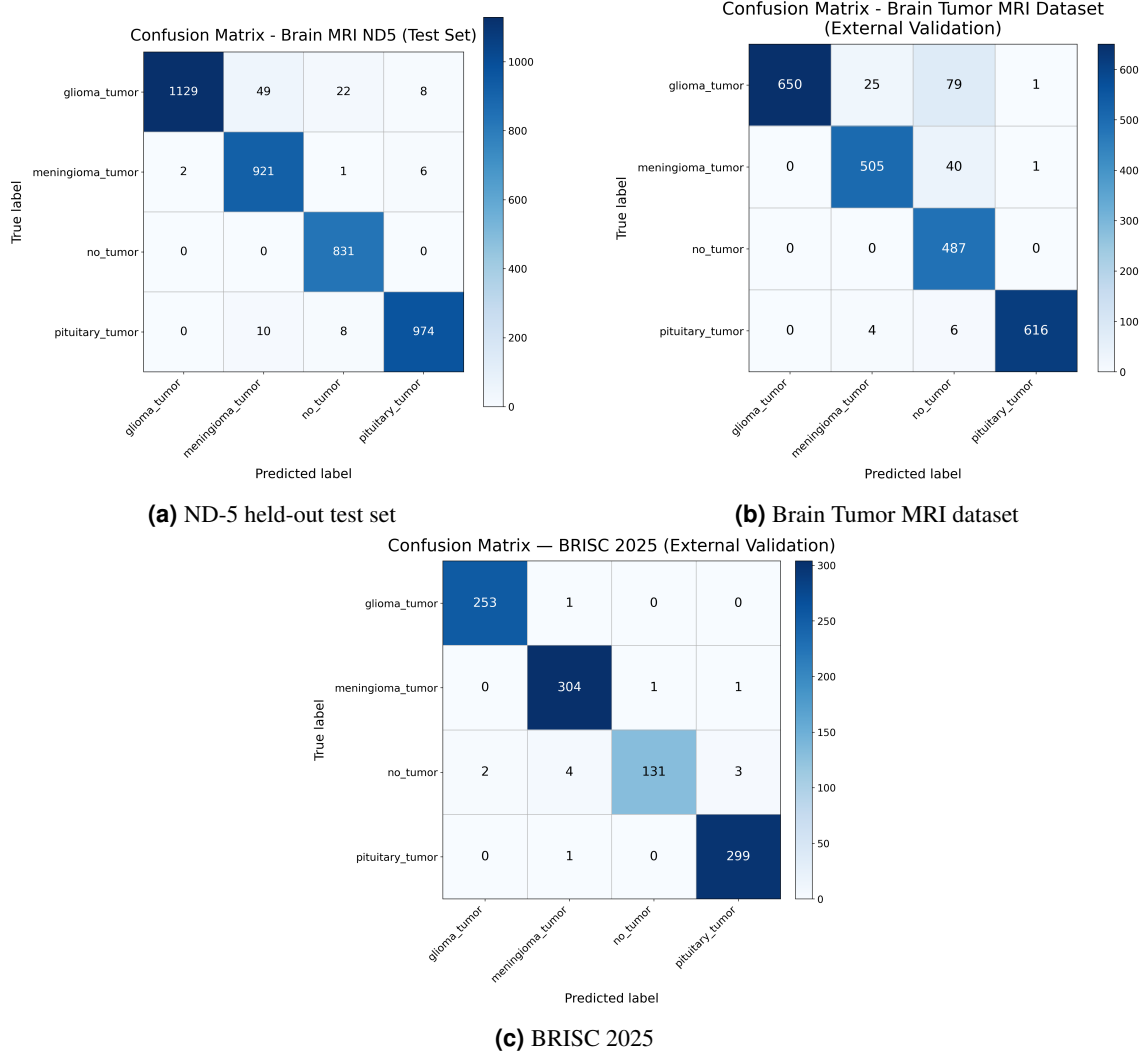


Figure 5. Confusion matrices for internal and external validation across datasets.

achieves a total accuracy of 97.32% on ND-5 with a steady high precision, recall, and F1-score of all four classes as shown in Table 3. `no_tumor` and `pituitary_tumor` performance is especially good since performance across both classes is high in terms of precision and recall and the discrimination between healthy scans and tumor ridden scans is also good and constant. The confusion matrix in Figure 5a also confirms this finding since most of the predictions are located along the diagonal. Mistakes are very few and limited within clinically plausible limits, the most common being of `glioma` which is mistaken with `meningioma` and in rare cases, `no_tumor`. It is interesting to note that `no_tumor` class is well maintained in this split indicating high degree of specificity in similar acquisition conditions.

External Validation

External validation on BRISC 2025 and Mendeley Brain Tumor MRI dataset shows generalization under distribution shift.. Table 4 shows 98.70% accuracy on BRISC 2025 and 93.54% accuracy on Tumor MRI Dataset of Mendeley, respectively with equal high macro precision, recall, F1-score, and AUC. Notably, these findings are achieved when the same deterministic settings are used on preprocessing and evaluation without dataset-specific tunings, thresholding, or augmentation alterations. As the confusion matrices in Figure 5c and Figure 5b demonstrate, the model is still extremely stable on the external data: BRISC 2025 has very few off-diagonal errors (Figure 5c), whereas the Brain Tumor MRI dataset of Mendeley shows relatively small rises in confusion between the tumor subclasses (Figure 5b), which is the expected response when the acquisition protocol and intensity distributions do not coincide with the fine-tuning domain. In general, it can be concluded that the outcomes represent that, when BraTS-motivated pretraining is merged with our attention-driven architecture, strong features are produced and transferred between datasets.

Table 3. Performance of Brain MRI ND-5 Test Dataset

Class	Precision (%)	Recall (%)	F1-score (%)
Glioma Tumor	100.00	91.89	95.77
Meningioma Tumor	92.43	99.78	95.97
No Tumor	96.06	99.76	97.88
Pituitary Tumor	98.78	97.98	98.38
Overall Accuracy (%)	97.32		

Table 4. External validation results on BRISC 2025 and Brain Tumor MRI Dataset from Mendeley.

Metric	BRISC 2025	Brain Tumor MRI Dataset
Accuracy (%)	98.70	93.54
Precision (%)	98.80	93.46
Recall (%)	98.05	94.25
F1-score (%)	98.40	93.43
AUC (%)	99.96	97.61

Representation Space and Optimization Analysis

Figure 6 shows the learned space as t-SNE embeddings of the ND-5 test set features. The representation has a good level of class-wise segregation but it also demonstrates intra-class organizational forms of variance. Specifically, the glioma class is split into two small sub clusters implying that the model represents different modes of glioma appearance that can occur due to variations in imaging view, acquisition, or tumour phenotype between slices. The fact that a few samples are at the border between glioma and meningioma shows the presence of partial similarities in features when dealing with cases that are visually ambiguous, which is similar with the remaining confusion between the two classes. In general, the embedding is well-separable between categories and boundary mixing is restricted to hard cases instead of broad overlap between classes. Such behavior of clustering is consistent with the diagonal dominance of confusion matrix in Figure 5a and indicates that the network acquires class-discriminative representations, and is not using dataset-specific shortcuts. Besides, training dynamics are consistent and in good generalization. In Figure 7a, the loss curve demonstrates that there is a continuous reduction in training loss and a constant validation loss, with no extreme deviation to signify very serious overfitting. Likewise, the accuracy curve in Figure 7b shows quick initial gains and saturation to a high validation accuracy near training curve, confirming the adaptability of the optimization strategy and regularization options.

Table 5. Accuracy comparison with prior work and the proposed Hi-TGNet, including model size.

Study/Method	Params (M)	Accuracy (%)
Gorenshtein et al. ⁴⁶	4.700	94.25
Saeedi et al. ⁴⁷	0.244	93.45
Dewage et al. ⁴⁸	16.87	94.51
Iftikhar et al. ⁴⁹	7.081	95.00
Hi-TGNet	3.981	ND-5 (in-domain test): 97.32
		BRISC 2025 (external): 98.70
		Brain Tumor MRI (external): 93.54

Comparison with Prior Work

Finally, we contextualize Hi-TGNet against recent published baselines in Table 5, reporting both accuracy and model size to reflect the accuracy efficiency trade-off. With 3.981M parameters, Hi-TGNet is more compact than models proposed by Dewage et al.⁴⁸, Iftikhar et al.⁴⁹, and Gorenshtein et al.⁴⁶, while achieving 97.32% accuracy on the ND-5 test split and retaining strong transfer under distribution shift on BRISC 2025 and the Brain Tumor MRI dataset. In contrast, the smallest model is by Saeedi et al.⁴⁷ with 0.244M parameters which reported a test accuracy of 93.45% after training on only 3,264 images that were expanded to 9,792 using simple rotation and flipping. Therefore, despite the attractive parameter count, the reliability of that result is limited without independent external validation.

Taken together, the in-domain and external results, the separable embedding structure, stable training dynamics, and the

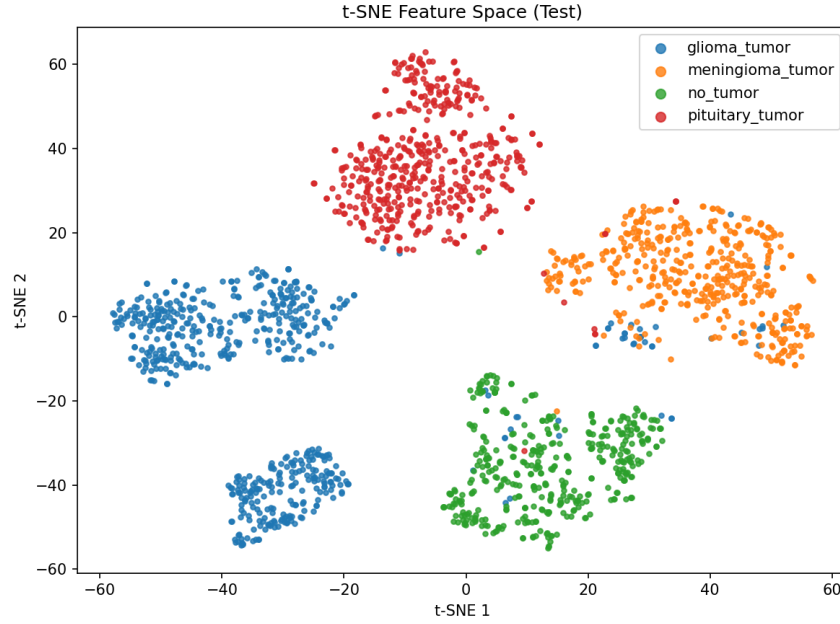


Figure 6. t-SNE visualization of the learned ND-5 test-set embeddings, showing class-wise feature separation in the representation space.

qualitative attribution evidence, supported by the accuracy and parameter profile in Table 5, indicate that Hi-TGNet offers a stronger balance of accuracy, model size, and out-of-distribution robustness. The key advantage is not only that Hi-TGNet remains lightweight, but that it is validated beyond a single, augmentation-amplified split, achieving 98.70% on BRISC 2025 and 93.54% on the Brain Tumor MRI dataset under the same deterministic inference protocol, which strengthens the practical significance of the comparison.

Explainability Analysis

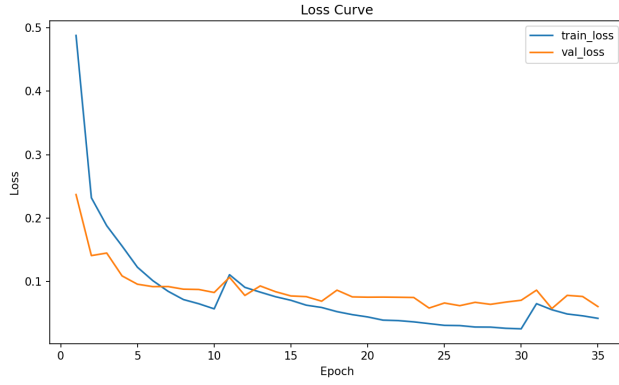
In addition to the quantitative performance, the qualitative visualizations offer more relevant details of the way the model makes its decisions. Tumor positive samples exhibit strong and spatially coherent activations in discrete and localized intracranial areas in Figure 8. In Sample 2 and Sample 3, Grad-CAM++ identifies generalized areas of interest, but with obvious locality, which coincide with abnormal tissue that could be seen. The overlay images demonstrate that the strongest responds are in the brain parenchyma and not along the skull border or edges of the image. The responses of Vanilla saliency are more scattered at the pixel level though the highest gradients are still concentrated to the same anatomical areas. Guided saliency further refined these responses and blocked background noises as well as gave more emphasis on the sharper structures within the suspected lesion region. The overlapping of the highlighted areas of all the three explanation systems enhance the argument that the network is actively interested in pathology relevant features.

Conversely, the `no_tumor` illustration in Sample 1 has weaker and diffused activations. The areas highlighted are not unified and have no strong center of focus. The given pattern indicates that the predication does not have a significant pathological signal. Generally, the cross method agreement and the spatial coincidence with realistic anatomical extensions show that the model is based on meaningful brain areas, as opposed to random artifacts. Such consistency between techniques of explanation also encourages the strength and clinical plausibility of the learned decision patterns.

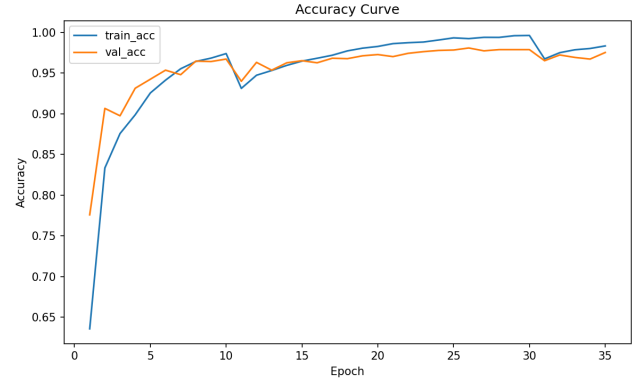
Key Findings From Our Investigation

We have found that the most effective results are obtained by setting pretraining, attention, regularization, and evaluation as a single pipeline addressing the primary failure modes of brain tumor classification. Most 2D classifiers fail due to the fact that they are susceptible to weak tumor localization, they fail in scanner and protocol shift condition, and they are overconfident in uncertain cases. As indicated in Table 3 and Table 4, our design can seal these gaps and generate good in-domain results on ND-5 and stable external results on BRISC 2025 and Brain Tumor MRI dataset from Mendeley.

One key discovery is that BraTS slice pretraining constructs a transferable prior tumor that ND-5 itself cannot deliver alone. ND-5 just feeds labels on classes and therefore a model can be pulled to dataset specific information. The encoder is trained on BraTS using three modalities as input channels where it is being trained to learn tumor sensitive features in a larger range of



(a) Loss curves over fine-tuning epochs on ND-5.



(b) Accuracy curves over fine-tuning epochs on ND-5.

Figure 7. Training dynamics during fine-tuning on the ND-5 dataset.

appearance variations. In pretraining, the model also learns to put spatial emphasis to tumor regions when mask guidance is on. This is part of the reason why the system was accurate under distribution shift in Table 4 instead of merely serving in-domain well.

We also conclude that multi scale modeling must be employed in separating tumor subclasses. The hierarchical attention modules and cross scale fusion retain the local evidence and the global context that enhance the separability of classes in the learned representation. This can be substantiated by the evident clustering in the t-SNE visualization in Figure 6 and the minimal off diagonal error in the confusion matrices, particularly on ND-5 in Figure 5a.

The other significant discovery is that it is controlled regularization and stable optimization, and not heavy augmentation that is required for generalization. The training curves become converged and the validation behavior can be observed without a large gap. This is apparent in Figure 7a and Figure 7b. The uncertainty aware head also encourages the use of disciplined predictions in cases of reduced evidence which aids in encouraging high performance in the external sets and reduction of reliance on spurious trends.

And, finally, the findings are valid due to rigorous and regular assessment. Validation and testing are carried out by means of deterministic preprocessing and predicting stabilization is achieved by means of test time augmentation. This reduces any variation to input perturbations and can be reproducible between datasets. Altogether, the overall effect is a model that is precise on ND-5 in Table 3, is robust on BRISC 2025 and Brain Tumor MRI Dataset of Mendeley in Table 4 and creates explainable focus regions that correspond to meaningful anatomy in Figure 8.

Summary of what worked best-

- BraTS slice pretraining learned robust tumor sensitive features which transferred to ND-5 and external datasets and enhanced stability.
- Subclass separability was enhanced by hierarchical attention as well as cross scale fusion.
- Regularization and uncertainty aware learning yielded stable convergence and avoid overfitting.
- Deterministic evaluation and test time augmentation enhanced reproducibility and minimized prediction variance across datasets.

Discussion

The findings indicate that a pretrained pipeline of BraTS and ND-5 fine-tuning is capable of obtaining high in-domain accuracy and at the same time ensure reliability on external validation. The results of the ND-5 test used in Table 3 and the confusion matrix in Figure 5a, suggest that the model is learning high levels of class level discrimination, with minor errors being made between the tumor subclasses and not between the tumor and the healthy cases. This is significant since it implies that the network is not merely identifying abnormality, but rather is acquiring the properties that can promote more discerning diagnosis. It is also supported by the fact that the learned representation can be transferred outside of ND-5 distribution, as evidenced by the external results in Table 4. The strong BRISC 2025 performance and the competitive Brain Tumor MRI Dataset performance are indicative that brain tumor pretraining stage assists to decrease the reliance on dataset particular textures and acquisition artifacts.

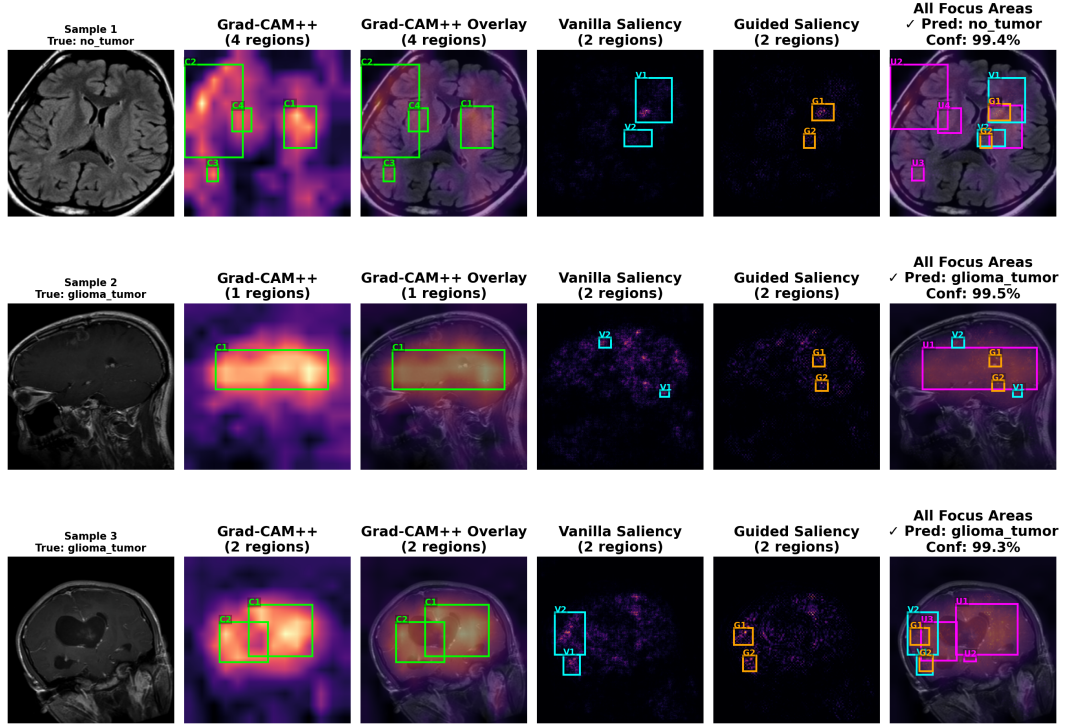


Figure 8. Qualitative explainability results using Grad-CAM++ and saliency-based methods, highlighting regions that contribute most to the model predictions.

A key takeaway is that robust performance appears to be driven by two complementary factors. First, slice level pretraining on BraTS trains a general tumor sensitive feature space, which is then exposed to ND-5 label set. Second, multi scale attention and fusion enhance the capability of disjunctly isolating delicate tumor classes by integrating the local evidence with the larger context. This is reflected in the apparent separation in the t-SNE plot in Figure 6, as the embedding clusters are in diagnostic categories, and exhibit little overlap. Moreover, the training process in Figure 7a and Figure 7b indicates that there is stable convergence and therefore, the augmentation, focal loss, attention regularization and uncertainty aware head combine to prevent overfitting. Lastly, the explainable component in Figure 8 in is a qualitative demonstration that the model tends to prioritize meaningful areas of tumor positive cases. Although attribution techniques are not evidence, the agreement of a number of explanations gives a certain degree of confidence that the decisions are not dominated by background artifacts.

Limitations and Future Scope

Despite the strong results, some limitations remain. First, the ND-5 data are 2D images and the model operates on single slices, so it cannot exploit full 3D context. This can be a source of confusion between tumor subtypes when discriminative cues appear only across adjacent slices. As a next step, we plan to extend the pipeline to 3D or 2.5D inputs, for example using short slice stacks, which can preserve computational efficiency while adding volumetric context. The uncertainty head provides a foundation, and we plan to quantify calibration using reliability curves and expected calibration error, and to investigate referral strategies that abstain on uncertain cases. Finally, we plan to ensure patient level splits when metadata are available in order to avoid potential leakage across correlated samples.

Conclusion

This study introduces a strong brain tumor classification pipeline, which uses slice level pretraining on BraTS 2020 dataset and fine tuning on Brain MRI ND-5 dataset to diagnose four classes. The method trains tumor sensitive representations that can be transferred across datasets and can achieve good in-domain performance without any drastic distribution shift. The multi scale attention and fusion design helps to achieve strong separation between categories of tumors and healthy cases, the uncertainty aware head promotes more disciplined predictions on vague samples. Training is steady with the convergence, and qualitative descriptions show that the model tends to be based on significant anatomical areas instead of artifacts in the background. All

these results indicate that tumor aware pretraining in combination with multi scale feature modeling and uncertainty guided learning is an effective and practical base towards reliable classification of brain tumors.

Data Availability

The datasets used in this study are publicly available from the following sources, subject to their respective usage terms and licenses. BraTS 2020 is available via the CBICA repository at <https://www.med.upenn.edu/cbica/brats2020/>. The Brain MRI ND-5 dataset is available through IEEE DataPort at <https://ieee-dataport.org/documents/brain-mri-nd-5-dataset>. External validation datasets include BRISC 2025, available on Kaggle at <https://www.kaggle.com/datasets/briscdataset/brisc2025/>, and the Brain Tumor MRI Dataset hosted on Mendeley Data at <https://data.mendeley.com/datasets/zwr4ntf94j/5>. The source code for Hi-TGNet is publicly available at https://github.com/Arche-Intelligence-Lab/brain_custom_cnn.

Author contributions

Conceptualization: S.R., M.N.S., T.M.J.; Methodology: S.R., M.N.S.; Software: S.R., M.N.S.; Validation: M.N.S., T.M.J.; Formal analysis: S.R., T.M.J.; Investigation: S.R., M.N.S., T.M.J.; Data curation: S.R., M.N.S., M.H.M.; Visualization: S.R.; Writing—original draft preparation: S.R., T.M.J.; Writing—review and editing: S.R., M.H.M., M.A.W., J.U.; Resources: M.N.S., T.M.J., M.A.W.; Supervision: M.I.M., J.U.; Project administration: M.I.M.; Funding acquisition: M.A.W.

All authors have read and approved the final published version of the manuscript.

Funding

This work was supported by Ongoing Research Funding Program - Research Chairs, King Saud University, Riyadh, Saudi Arabia, under Grant ORF-RC-2026-2000.

References

1. Kim, S. *et al.* Global burden of brain and central nervous system cancer in 185 countries, and projections up to 2050: a population-based systematic analysis of globocan 2022. *J. Neuro-Oncology* **175**, 673–685 (2025).
2. Ostrom, Q. T. *et al.* Cbtrus statistical report: pediatric brain tumor foundation childhood and adolescent primary brain and other central nervous system tumors diagnosed in the united states in 2014–2018. *Neuro-oncology* **24**, iii1–iii38 (2022).
3. Verma, S., Malviya, R. & Uniyal, P. Survival of patients with primary brain tumor: A data analysis of 10 years (2024).
4. Chiu, F.-Y. & Yen, Y. Imaging biomarkers for clinical applications in neuro-oncology: current status and future perspectives. *Biomark. Res.* **11**, 35 (2023).
5. Obuchowicz, R. *et al.* Artificial intelligence-empowered radiology—current status and critical review. *Diagnostics* **15**, 282 (2025).
6. Bouhafra, S. & El Bahi, H. Deep learning approaches for brain tumor detection and classification using mri images (2020 to 2024): a systematic review. *J. Imaging Informatics Medicine* **38**, 1403–1433 (2025).
7. Haque, M. E., Nurul Absur, M., Al Farid, F., Uddin, J. & Abdul Karim, H. A novel interpretable and real-time dengue prediction framework using clinical blood parameters with genetic and gan-based optimization. *Front. Artif. Intell.* **Volume 8 - 2025**, DOI: [10.3389/frai.2025.1626699](https://doi.org/10.3389/frai.2025.1626699) (2025).
8. Li, M., Jiang, Y., Zhang, Y. & Zhu, H. Medical image analysis using deep learning algorithms. *Front. public health* **11**, 1273253 (2023).
9. Banik, S., Das, M., Banik, T. & Haque, M. E. Mobiledenseattn:a dual-stream architecture for accurate and interpretable brain tumor detection (2025). [2508.18294](https://doi.org/10.26434/chemrxiv-2025-2508).
10. Vamsidhar, D. *et al.* Hybrid model integration with explainable ai for brain tumor diagnosis: a unified approach to mri analysis and prediction. *Sci. Reports* **15**, 20542 (2025).
11. Chinga, A., Bendezu, W. & Angulo, A. Comparative study of cnn architectures for brain tumor classification using mri: Exploring gradcam for visualizing cnn focus. *Eng. Proc.* **83**, 22 (2025).
12. Asiri, A. A. *et al.* Advancing brain tumor classification through fine-tuned vision transformers: A comparative study of pre-trained models. *Sensors* **23**, 7913 (2023).

13. Tandel, G. S., Tiwari, A. & Kakde, O. G. Multi-class brain tumor grades classification using a deep learning-based majority voting algorithm and its validation using explainable-ai. *J. Imaging Informatics Medicine* 1–38 (2025).
14. Tian, Y. Trade-off analysis of classical machine learning and deep learning models for robust brain tumor detection: Benchmark study. *JMIR AI* **4**, e76344 (2025).
15. Bhagyalaxmi, K., Dwarakanath, B. & Reddy, P. V. P. Deep learning for multi-grade brain tumor detection and classification: a prospective survey. *Multimed. Tools Appl.* **83**, 65889–65911 (2024).
16. Gocer, E. Medical image data augmentation: techniques, comparisons and interpretations. *Artif. intelligence review* **56**, 12561–12605 (2023).
17. Rahman, T., Islam, M. S. & Uddin, J. Mri-based brain tumor classification using a dilated parallel deep convolutional neural network. *Digital* **4**, 529–554 (2024).
18. Kong, G., Wu, C., Zhang, Z., Yin, C. & Qin, D. M 3: using mask-attention and multi-scale for multi-modal brain mri classification. *Front. Neuroinformatics* **18**, 1403732 (2024).
19. Alshomrani, F. Challenges and advances in classifying brain tumors: An overview of machine, deep learning, and hybrid approaches with future perspectives in medical imaging. *Curr. Med. Imaging* **21**, E15734056365191 (2025).
20. Puttegowda, K., Govindgowda, M., Mayigegowda, P., Ramegowda, P. & Nagaraju, A. M. Automated brain tumor detection with advanced machine learning techniques. *Biomed. Pharmacol. J.* **18** (2025).
21. Dhole, N. V., Dixit, V. V. & Mahajan, R. G. A review of deep learning and machine learning techniques for brain and breast cancer detection: Challenges and future directions. *J. Mech. Medicine Biol.* 2530002 (2025).
22. Chen, Y. *et al.* A novel deep learning method for brain tumor segmentation in magnetic resonance images based on residual units and modified u-net model. *J. Mech. Medicine Biol.* **23**, 2340088 (2023).
23. Thahiruddin, M. & Wulandari, A. Cnns vs. hybrid transformers for brain tumor classification on the brisc dataset. *J. Appl. Technol. Inf. Manag. (JATIM)* **6** (2025).
24. Chandraprabha, K., Ganesan, L. & Baskaran, K. A novel approach for the detection of brain tumor and its classification via end-to-end vision transformer-cnn architecture. *Front. Oncol.* **15**, 1508451 (2025).
25. Cakmak, Y. & Pacal, I. Comparative analysis of transformer architectures for brain tumor classification. *Explor. Medicine* **6**, 1001377 (2025).
26. Tariq, A., Iqbal, M. M., Iqbal, M. J. & Ahmad, I. Transforming brain tumor detection empowering multi-class classification with vision transformers and efficientnetv2. *IEEE Access* (2025).
27. Zhang, W., Chen, S., Ma, Y., Liu, Y. & Cao, X. Etunet: Exploring efficient transformer enhanced unet for 3d brain tumor segmentation. *Comput. Biol. Medicine* **171**, 108005 (2024).
28. Safwan, M. N. *et al.* T3sslnet: triple-method self-supervised learning for enhanced brain tumor classification in mri. *IEEE Access* (2025).
29. Latif, S. *et al.* Explainable ai in medical imaging: an interpretable and collaborative federated learning model for brain tumor classification. *Front. Oncol.* **15**, 1535478 (2025).
30. Gundogan, E. A novel hybrid deep learning model enhanced with explainable ai for brain tumor multi-classification from mri images. *Appl. Sci.* **15**, 5412 (2025).
31. Ma, Y. Federated learning for brain tumor diagnosis: Methods, challenges and future prospects. In *ITM Web of Conferences*, vol. 70, 03028 (EDP Sciences, 2025).
32. Sheikholeslami, S. *et al.* Utilizing large language models for ablation studies in machine learning and deep learning. In *Proceedings of the 5th Workshop on Machine Learning and Systems*, 230–237 (2025).
33. Prokopiou, I. & Spyridonos, P. Highlighting the advanced capabilities and the computational efficiency of deeplabv3+ in medical image segmentation: An ablation study. *BioMedInformatics* **5**, 10 (2025).
34. Salam, A. *et al.* Channel-attention densenet with dilated convolutions for mri brain tumor classification. *Comput. Model. Eng. & Sci.* **145**, 2457 (2025).
35. Al-Razgan, M., Ali, Y. A. & Awwad, E. M. Enhancing fetal medical image analysis through attention-guided convolution: a comparative study with established models. *J. Disabil. Res.* **3**, 20240005 (2024).
36. Abdusalomov, A. B., Mukhiddinov, M. & Whangbo, T. K. Brain tumor detection based on deep learning approaches and magnetic resonance imaging. *Cancers* **15**, 4172 (2023).

37. Hayat, M. T. *et al.* A hybrid convolutional neural network–long short-term memory (cnn–lstm)–attention model architecture for precise medical image analysis and disease diagnosis. *Diagnostics* **15**, 2673 (2025).
38. Disci, R., Gurcan, F. & Soylu, A. Advanced brain tumor classification in mr images using transfer learning and pre-trained deep cnn models. *Cancers* **17**, 121 (2025).
39. He, L., Luan, L. & Hu, D. Deep learning-based image classification for ai-assisted integration of pathology and radiology in medical imaging. *Front. Medicine* **12**, 1574514 (2025).
40. Bakas, S. *et al.* Advancing the cancer genome atlas glioma mri collections with expert segmentation labels and radiomic features. *Sci. data* **4**, 1–13 (2017).
41. Bakas, S. *et al.* Identifying the best machine learning algorithms for brain tumor segmentation, progression assessment, and overall survival prediction in the brats challenge. *arXiv preprint arXiv:1811.02629* (2018).
42. Menze, B. H. *et al.* The multimodal brain tumor image segmentation benchmark (brats). *IEEE transactions on medical imaging* **34**, 1993–2024 (2014).
43. Safwan, M. N., Rahman, S., Mahadi, M. H., Jabir, T. M. & Mobin, I. Brain mri nd-5 dataset, DOI: [10.21227/q2vt-tf46](https://doi.org/10.21227/q2vt-tf46) (2024).
44. Fateh, A. *et al.* Brisc: Annotated dataset for brain tumor segmentation and classification with swin-hafnet. *arXiv preprint arXiv:2506.14318* (2025).
45. Hira, M. I. K. *et al.* Brain tumor MRI dataset (glioma, meningioma, pituitary, no tumor), DOI: [10.17632/zwr4ntf94j.5](https://doi.org/10.17632/zwr4ntf94j.5) (2025).
46. Gorenstein, A., Liba, T. & Goren, A. Lightweight transfer learning models for multi-class brain tumor classification: Glioma, meningioma, pituitary tumors, and no tumor mri screening. *J. Imaging Informatics Medicine* 1–10 (2025).
47. Saeedi, S., Rezayi, S., Keshavarz, H. & R. Niakan Kalhori, S. Mri-based brain tumor detection using convolutional deep learning methods and chosen machine learning techniques. *BMC Med. Informatics Decis. Mak.* **23**, 16 (2023).
48. Dewage, K. A. K. W., Hasan, R., Rehman, B. & Mahmood, S. Enhancing brain tumor detection through custom convolutional neural networks and interpretability-driven analysis. *Information* **15**, 653 (2024).
49. Iftikhar, S., Anjum, N., Siddiqui, A. B., Ur Rehman, M. & Ramzan, N. Explainable cnn for brain tumor detection and classification through xai based key features identification. *Brain Informatics* **12**, 10 (2025).